LifeLink - TODO / Backlog

D3, version 1 | Updated 22 October 2026 | Due 23 October 2026, 23:59 Türkiye time

CMPE 491 | İlayda Tsetsekos, Mehmet Alp Vural, Alaettin Doğan

Project website: https://iaaproject.github.io/pproject/

Progress summary

Sprint	Tasks	Complete	Points	Points complete	% points complete
Sprint 1 (6-16 Oct)	9	9	38	38	100%
Sprint 2 (19-23 Oct)	16	8	34	26	76%
Sprint 3 (26 Oct-20 Nov)	15	0	53	0	0%
Sprint 4 (23 Nov-31 Dec)	13	0	53	0	0%
Sprint 5 (Jan 2027)	3	0	11	0	0%
Total	56	17	189	64	34%

Status and planning conventions

Complete means the named task is finished. Writing completion does not include team approval, LMS submission or website publication; these have separate tasks.

Team denotes shared responsibility for completed document preparation. Initial D2 and D3 drafts prepared by İlayda Tsetsekos; team review pending. Other individual responsibilities are retained where already established. Unconfirmed future owners remain To be agreed.

Story points estimate relative effort from 1 to 8, not hours or elapsed progress. Split tasks are estimated separately, so the total differs from the earlier draft. Summary counts are recalculated from the task rows.

Story identifies a user-facing story rather than a general task. Sprint Ready indicates whether a task is prepared for near-term planning, not whether it is finished. Future due dates marked TBA have not yet been agreed.

Both the wearable flow and the separate offline messaging flow belong to the first demonstration. Initial bench power is USB. LoRa is preferred for the wearable link; Wi-Fi/BLE may be a demonstration fallback. Final interface, hardware and battery choices remain open.

Sprint 1 (6-16 Oct)

Initial demonstration and report preparation

ID	Task and checks	Area	Prio.	Status	SP	Owner	Due	D#
1.1	**Select project topic and title** Story: No; Sprint Ready: Yes	Project	High	Complete	3	Team	2026-10-12	D1
1.2	**Secure supervisor and jury members** Story: No; Sprint Ready: Yes Project information sent to Seydim; D1 sent to supervisor and jury on 16 October.	Project	High	Complete	2	Team	2026-10-13	D1
1.3	**Create project website with Reports section** Story: No; Sprint Ready: Yes	Website	High	Complete	5	Alaettin Doğan	2026-10-15	D1
1.4	**Write D1 proposal and initial literature review** Story: No; Sprint Ready: Yes Initial proposal completed; submitted copy retained.	Reports	High	Complete	8	Team	2026-10-16	D1
1.5	**Submit D1 to LMS** Story: No; Sprint Ready: Yes Submitted 16 October, 17:10.	Reports	High	Complete	1	İlayda Tsetsekos	2026-10-16	D1
1.6	**Email D1 to supervisor and jury** Story: No; Sprint Ready: Yes	Reports	Medium	Complete	1	Mehmet Alp Vural	2026-10-16	D1
1.7	**Prepare wearable design decisions working document** Story: No; Sprint Ready: Yes Design planning completed; hardware implementation is tracked separately.	Wearable	Medium	Complete	8	Team	2026-10-14	-
1.8	**Prepare initial quarantine working plan** Story: No; Sprint Ready: Yes Open for further team development.	Security	Medium	Complete	5	Team	2026-10-14	-
1.9	**Prepare TinyML data, labelling and evaluation plan** Story: No; Sprint Ready: Yes Use the latest maintained plan. This records planning, not a trained or deployed model.	AI	Medium	Complete	5	Team	2026-10-15	-

Sprint 2 (19-23 Oct)

Initial demonstration and report preparation

ID	Task and checks	Area	Prio.	Status	SP	Owner	Due	D#
2.1	**Agree team role assignments** Story: Yes; Sprint Ready: Yes Confirm owners together; unconfirmed assignments remain To be agreed.	Project	High	In Progress	1	Team	2026-10-23	D3
2.2	**Compare messaging alternatives and document selection criteria** Story: No; Sprint Ready: Yes Local Wi-Fi browser page versus separate radio interface. Final selection is task 3.14.	Messaging	High	Complete	3	Team	2026-10-22	D2
2.3	**Ask supervisor about ethics scope and laboratory inventory; confirm report-submission instructions** Story: No; Sprint Ready: Yes No laboratory stock or ethics permission is assumed. Track requests and replies separately.	Project	High	Not Started	1	To be agreed	2026-10-23	-
2.4	**Write D2 introduction and description** Story: No; Sprint Ready: Yes Writing completed; team review is separate in 2.9b.	Reports	High	Complete	3	Team	2026-10-22	D2
2.5	**Write D2 constraints** Story: No; Sprint Ready: Yes Writing completed; team review is separate in 2.9b.	Reports	High	Complete	5	Team	2026-10-22	D2
2.6	**Write D2 professional and ethical responsibilities** Story: No; Sprint Ready: Yes Writing completed; team review is separate in 2.9b.	Reports	High	Complete	3	Team	2026-10-22	D2
2.7	**Write D2 functional and nonfunctional requirements** Story: No; Sprint Ready: Yes Writing completed; team review is separate in 2.9b.	Reports	High	Complete	5	Team	2026-10-22	D2
2.8	**Extend and verify the D2 literature review and platform comparison** Story: No; Sprint Ready: Yes Writing completed; team review is separate in 2.9b.	Reports	High	Complete	3	Team	2026-10-22	D2
2.9a	**Assemble D2 and check spelling, consistency and PDF layout** Story: No; Sprint Ready: Yes	Reports	High	Complete	2	Team	2026-10-22	D2

ID	Task and checks	Area	Prio.	Status	SP	Owner	Due	D#
2.9b	**Review and agree D2 as a team** Story: No; Sprint Ready: Yes Review open decisions and requirements before submission.	Reports	High	Not Started	1	To be agreed	2026-10-23	D2
2.9c	**Upload the final D2 PDF to LMS** Story: No; Sprint Ready: Yes Deadline 23:59 Türkiye time. Retain submission confirmation.	Reports	High	Not Started	1	To be agreed	2026-10-23	D2
2.9d	**Publish the final D2 PDF on the project website** Story: No; Sprint Ready: Yes Check the published link opens the submitted version.	Website	High	Not Started	1	To be agreed	2026-10-23	D2
2.10a	**Update backlog v1 and produce a readable PDF** Story: No; Sprint Ready: Yes This updated document; team review and submission remain open.	Project	High	Complete	2	Team	2026-10-22	D3
2.10b	**Review backlog owners, estimates and status as a team** Story: No; Sprint Ready: Yes Confirm rather than infer individual responsibility.	Project	High	Not Started	1	To be agreed	2026-10-23	D3
2.10c	**Upload the final D3 PDF to LMS** Story: No; Sprint Ready: Yes Deadline 23:59 Türkiye time. Retain submission confirmation.	Project	High	Not Started	1	To be agreed	2026-10-23	D3
2.10d	**Publish backlog v1 on the project website** Story: No; Sprint Ready: Yes Use the existing D3 section and verify the PDF link.	Website	High	Not Started	1	To be agreed	2026-10-23	D3

Sprint 3 (26 Oct-20 Nov)

Initial demonstration and report preparation

ID	Task and checks	Area	Prio.	Status	SP	Owner	Due	D#
3.1	**D4: scenarios and use-case model** Story: Yes; Sprint Ready: No	Reports	High	Not Started	5	To be agreed	2026-11-20	D4
3.2	**D4: object/class and dynamic models** Story: Yes; Sprint Ready: No	Reports	High	Not Started	5	To be agreed	2026-11-20	D4
3.3	**D4: dashboard and messaging-interface UI mock-ups** Story: Yes; Sprint Ready: No Keep messaging mock-ups conditional on task 3.14.	Reports	Medium	Not Started	3	To be agreed	2026-11-20	D4
3.4	**Submit an ethics application if required; obtain permission before relevant human tests** Story: No; Sprint Ready: No Depends on 2.3. Public-data work and non-human bench tests can continue while approval is pending.	Project	High	Not Started	3	To be agreed	TBA	-
3.5	**Select and source stage-1 MCU and sensor modules** Story: No; Sprint Ready: No Check 2.3 inventory first. MAX30101 and ADXL362/LIS2DH12 are candidates. Compare drivers, cost and availability before purchase.	Wearable	High	Not Started	2	To be agreed	TBA	-
3.6	**Log timestamped sensor samples in USB-powered non-human bench tests** Story: No; Sprint Ready: No Check sample counts, timing and storage. Bench USB operation does not establish battery runtime.	Wearable	Medium	Not Started	5	To be agreed	TBA	-
3.7	**Select the wearable-gateway link and source compatible boards** Story: No; Sprint Ready: No LoRa preferred; Wi-Fi/BLE may support the first demonstration. Decide from inventory, cost, two-way operation and setup effort. Independent of messaging interface 3.14.	Wearable	High	Not Started	3	To be agreed	TBA	-
3.8	**Audit PPG-DaLiA and align windows with ECG-derived references** Story: No; Sprint Ready: No Record dataset/version and invalid-reference handling; preserve participant separation.	AI	Medium	Not Started	5	To be agreed	TBA	-

ID	Task and checks	Area	Prio.	Status	SP	Owner	Due	D#
3.9	**Implement fixed v0 pulse estimator and generate reference-based labels** Story: No; Sprint Ready: No Freeze estimator, log-amplitude interpolation guards and error tolerance before outcome-based tuning. ECG is not a model input.	AI	Medium	Not Started	5	To be agreed	TBA	-
3.10	**Implement rule-based quality baseline and observable reason codes** Story: No; Sprint Ready: No Separate uncertain measurements from accepted estimates. No diagnostic absence-of-life labels.	AI	Medium	Not Started	5	To be agreed	TBA	-
3.11	**Create removable capsule and strap mock-ups** Story: No; Sprint Ready: No Use laboratory 3D printing if available. Human comfort evaluation depends on the applicable ethics process.	Wearable	Low	Not Started	3	To be agreed	TBA	-
3.12	**Prepare wristband concept drawing or logo for the website** Story: No; Sprint Ready: No Volunteered.	Website	Low	Not Started	2	İlayda Tsetsekos	TBA	-
3.13	**D5: update backlog v2, upload to LMS and publish** Story: No; Sprint Ready: No	Project	High	Not Started	1	To be agreed	2026-11-20	D5
3.14	**Select the offline messaging interface** Story: No; Sprint Ready: No Choose local Wi-Fi browser page or separate radio interface using 2.2 criteria. Do not infer this choice from the wearable radio.	Messaging	High	Not Started	3	To be agreed	TBA	-
3.15	**D4: freeze test thresholds, timing budgets and exceptional-state rules** Story: No; Sprint Ready: No Define freshness, airtime/retries, manual-help limits, command expiry and listening windows; trace to D2 requirements.	Reports	High	Not Started	3	To be agreed	2026-11-20	D4

Sprint 4 (23 Nov-31 Dec)

Initial demonstration and report preparation

ID	Task and checks	Area	Prio.	Status	SP	Owner	Due	D#
4.1	**D6: subsystem decomposition and deployment diagram** Story: Yes; Sprint Ready: No	Reports	High	Not Started	5	To be agreed	2026-12-31	D6
4.2	**D6: security design for authenticated temporary identifiers and encrypted reports** Story: Yes; Sprint Ready: No Specify keys, AES-CCM candidate, associated data, nonce/counter uniqueness, restart handling and duplicate-command exception.	Security	High	Not Started	5	To be agreed	2026-12-31	D6
4.3	**D6: boundary conditions and global control** Story: Yes; Sprint Ready: No	Reports	Medium	Not Started	3	To be agreed	2026-12-31	D6
4.4	**Integrate button, report, gateway, dashboard and operator acknowledgement** Story: Yes; Sprint Ready: No Keep local storage, gateway receipt and operator acknowledgement distinct. Confirm only received operator acknowledgement by vibration; LED is secondary.	Integration	High	Not Started	8	To be agreed	TBA	D8
4.5a	**Implement bounded routine-report pause and automatic expiry** Story: Yes; Sprint Ready: No Manual help remains independently eligible under its own small budget and applicable airtime limits. Test help during the pause.	Security	High	Not Started	3	To be agreed	TBA	D8
4.5b	**Test independent listening and duplicate-command acknowledgement recovery** Story: Yes; Sprint Ready: No Drop the first application acknowledgement. Resend the identical command through an independent window without button input; return status without restarting the timer.	Security	High	Not Started	3	To be agreed	TBA	D8
4.5c	**Test invalid commands and truthful restriction display** Story: Yes; Sprint Ready: No Reject wrong authentication tags and older counters. Display queued/unconfirmed/applied states; planned silence, ageing measurement and approximate return time from remaining duration.	Security	High	Not Started	3	To be agreed	TBA	D8

ID	Task and checks	Area	Prio.	Status	SP	Owner	Due	D#
4.6a	**Implement offline help-message and operator-reply flow** Story: Yes; Sprint Ready: No Use the interface selected in 3.14; show it beside wearable reports. Disable internet during the demonstration.	Messaging	High	Not Started	5	To be agreed	TBA	D8
4.6b	**Implement bounded messaging admission and retry eligibility** Story: Yes; Sprint Ready: No Per-session plus aggregate limits. A limited session retains a later help opportunity. If browser-based, do not rely only on a resettable cookie.	Security	High	Not Started	3	To be agreed	TBA	D8
4.7	**Train and evaluate initial feature-based quality models on a computer** Story: No; Sprint Ready: No Nested participant-separated LOSO; compare no screening, acceleration threshold, rules, logistic regression and small MLP. Report error-coverage, subjects and activities; no accepted windows means undefined accepted-window error.	AI	Medium	Not Started	8	To be agreed	TBA	-
4.8	**Measure whole-board power and build an airtime/energy budget** Story: No; Sprint Ready: No Include standby, sensing, processing, TX, RX, independent listening and feedback. State payload/radio settings and gateway downlink budget. Separate measured board results from estimates.	Wearable	Medium	Not Started	3	To be agreed	TBA	-
4.9	**D7: update backlog v3, upload to LMS and publish** Story: No; Sprint Ready: No	Project	High	Not Started	1	To be agreed	2026-12-31	D7
4.10	**Select a protected battery and prepare portable power integration** Story: No; Sprint Ready: No Use 4.8 measurements, pulse-current capability, protection and enclosure dimensions. Initial bench work uses USB. Long-term use without routine recharge remains an evaluated goal.	Wearable	Medium	Not Started	3	To be agreed	TBA	-

Sprint 5 (Jan 2027)

Poster, presentation and initial demonstration

ID	Task and checks	Area	Prio.	Status	SP	Owner	Due	D#
5.1	**Prepare poster** Story: No; Sprint Ready: No January demonstration date to be announced.	Presentation	High	Not Started	3	To be agreed	TBA	D8
5.2	**Prepare the 25-minute recorded presentation with a speaking part for every member** Story: No; Sprint Ready: No Plan for the course presentation format; confirm the final timing instructions.	Presentation	High	Not Started	5	To be agreed	TBA	D8
5.3	**Rehearse both demonstration flows and prepare a labelled recorded-data fallback** Story: No; Sprint Ready: No Exercise loss, uncertain measurement, help during restriction and offline replies. Live human measurements require the applicable permission.	Integration	High	Not Started	3	To be agreed	TBA	D8

Second-semester scope

On-device quality deployment, multi-device traffic and obstruction experiments, automatic restriction with a recovery waiting period, command replacement/early release, and optional field relays remain second-semester work. Compare the same traffic scenario with control enabled and disabled. These extensions are described in D2 and are not marked as completed first-semester implementation tasks.

For D3, D5 and D7: review current task rows, export a readable PDF, upload it to LMS and publish the submitted version on the project website. Update status only after each action is complete.